PREDICT NEW YORK CITY AIRBNB PRICES

Linear Regression — One-Page Summary Report

Project Objective

The objective is to estimate nightly Airbnb listing prices in New York City using a Linear Regression model. The project instructions specify exploratory data analysis, data cleaning, outlier treatment, categorical encoding, an 80/20 train-test split, model evaluation, and residual analysis.

Results from the Provided Prediction File

The supplied ZIP file contains 9,150 rows with actual_price, predicted_price, and residual values. The following metrics were calculated by comparing actual and predicted prices in that file.

Metric	Result
R-squared (R^2)	0.487 (48.7%)
Mean Absolute Error (MAE)	$38.32 per listing
Mean Squared Error (MSE)	2,760.62 dollars squared
Root Mean Squared Error (RMSE)	$52.54

What the Metrics Mean

The R^2 score is 0.487, meaning the predictions account for about 48.7% of the variation in actual prices relative to a simple mean-price baseline. The MAE is $38.32, so a prediction differs from the actual nightly price by about this amount on average. The RMSE is $52.54; it penalizes larger errors more strongly than MAE.

Cheap vs. Expensive Listings

Using the median actual price ($100) to split the supplied results, listings at or below the median had an MAE of $29.01 (n=4,708), while listings above the median had an MAE of $48.18 (n=4,442). The model made smaller average dollar errors on cheaper listings. More expensive listings may be harder to estimate because prices vary widely and unusual or luxury properties can have different pricing patterns.

Conclusion and Limitation

On the provided prediction results, the model shows a moderate fit and an average absolute error of about $38 per night. The larger MAE for above-median listings suggests expensive listings are harder to predict accurately. Limitation: the uploaded ZIP contains prediction results only, not the original raw Airbnb dataset or the full notebook/code. This report evaluates the supplied predictions but cannot independently verify the EDA, cleaning, feature encoding, or train/test split required by the project instructions.